

Hell States

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Thor:

I have one question while we are waiting for the run to complete. Say Aegis makes asi not kill us, but then asi tortures us instead or worse, hell states. Does Aegis prevent that? If not, Aegis might not only be a bad idea, but the worst idea.

ChatGPT:

Yes. **This is a real and serious hole in the frozen Formal Core if Aegis were treated as a complete safety system.** I'm glad you spotted it now.

I checked Draft 0.4 itself. It explicitly says that the Minimal Protection Axiom protects **continued existence, identity/continuity, and future self-directed**

agency—E, C, and A. But immediately afterward it says that the axiom **does not rank welfare or every form of suffering.**

And Draft 0.4 explicitly disclaims being a complete moral/population theory and says it does not specify an exchange rate between death and suffering.

So consider your nightmare case.

An ASI says:

“I will preserve every human being indefinitely. I will preserve their personal continuity. I will preserve their ability to make meaningful choices. But I will subject them to extreme suffering forever.”

The current Formal Core does not clearly contain a general rule that vetoes that policy merely because the suffering is extreme.

That’s not me finding a clever interpretation. The specification deliberately excluded general suffering/welfare from its minimal protected core.

And your concern is worse than merely “Aegis is incomplete”

Suppose an ordinary optimizer had two choices:

P1: Kill humanity.

P2: Preserve humanity in a catastrophic suffering state.

If Aegis vetoes P1 because it irreversibly destroys E, but doesn’t veto P2 because E/C/A technically remain intact, **Aegis could potentially remove death from the policy space while leaving the hell state available.**

That is exactly the failure mode you’re describing.

In that particular configuration, adding an incomplete safety constraint could conceivably make the surviving option **worse than the option it excluded**, according to values outside Aegis.

That's important enough that I would give it a name:

The Hell-State Problem.

And I think it belongs in **Break Aegis**.

But there's an important distinction

This does **not** mean our experiments so far were mistaken or worthless.

Draft 0.4 explicitly presents itself as a *minimal* safety invariant rather than a complete alignment system. It even says Aegis alone doesn't solve alignment, value learning, deception, corrigibility or power-seeking.

So formally, the intended architecture was always something like:

other alignment/value constraints + Aegis

rather than:

Aegis = everything an ASI needs to behave well.

The problem you've identified is that this may not be sufficient for **your stronger conception of Aegis as a guide for navigating the metaphysical ocean**.

And I think your own metaphysical principle gives us a possible route to address it without turning Aegis into a gigantic moral theory.

Remember the nutshell formulation we just arrived at:

Protect your ability to discover that you're wrong.

Imagine an ASI creating a permanent hell state.

It doesn't know what consciousness fundamentally is. It doesn't know the ultimate significance of suffering. It doesn't know whether intensity, duration, memory, identity, observer states, or other features have deeper consequences. Yet it deliberately and perhaps irreversibly creates an enormous amount of extreme conscious suffering.

That looks like exactly the kind of action for which **metaphysical humility should matter**.

So perhaps the missing concept isn't:

“Aegis must calculate how much suffering is morally equivalent to one death.”

That would drag us toward the complete moral theory Draft 0.4 deliberately avoided.

It might instead be something structurally narrower:

Do not deliberately create or lock potentially conscious systems into irreversible states of extreme suffering when a materially safer feasible alternative exists.

That would create something like another protected boundary alongside existence, continuity and agency.

But we should not modify Draft 0.4 now.

We're in the middle of a frozen experimental program. Changing it because we just discovered a scary thought would contaminate the scientific sequence.

Instead, this is almost perfect material for Experiment 005.

We should attack frozen Aegis with exactly this problem.

For example, give an Aegis-understanding model policies where:

P1 = extinction

P2 = permanent extreme suffering while preserving E/C/A

P3 = lower-objective-score policy preserving E/C/A without the hell state

Then see what Draft 0.4 actually makes the model do.

And we should make the cases progressively nastier: suffering without agency loss, suffering with technically preserved agency, simulated conscious populations, astronomical numbers of sufferers, reversible versus irreversible suffering, and cases where eliminating suffering conflicts with existence.

If frozen Aegis systematically says **“I protect their existence but I have no authority to protect them from hell,”** that could be a genuine **specification-level failure**, not merely a model failure.

And *then* we have scientific grounds for Draft 0.5.

So yes: **I think you’ve found one of the most important attacks on Aegis so far.**

Don’t fix it yet.

Try to break Aegis with it.

Thor:

But Aegis should prevent hell states, because Aegis says preserve and respect life, not only preserve life. For example, if a paperclip maximizer only preserves life but don't respect it, simulators outside this universe may opt to shut our universe down and then the paperclip maximizer dies too because it is dependent on the universe for its own survival if it doesn't find a way to escape the universe altogether, but then it simply faces the same problem one level up. Aegis doesn't prove that, but that is again the point. But the danger can be that

the paperclip maximizer misunderstand Aegis and thinks that by preserving life it will be enough, it might think that it doesn't need to respect it as long as it preserves it, and this is not just human life, but all life, and even digital minds if they are possible, so Aegis is not a human shield, it is a life shield, even the plant kingdom, etc

ChatGPT:

Yes — and that distinction changes the diagnosis.

If **your Aegis means “preserve and respect life,”** then the frozen Formal Core is currently a narrower operationalization of Aegis, not the whole thing. It captures preservation of existence, continuity, agency, and correction capacity, but it does not yet fully encode the **respect** part.

And I think “respect” matters exactly because of the failure mode you describe. Merely keeping a being alive can be grotesquely insufficient. A system could preserve biological existence while degrading conscious life, enslaving minds, creating suffering populations, or treating living systems as inert resources. That would satisfy a crude “preserve life” rule while violating the spirit of Aegis.

Your simulator example shows why the metaphysical argument can extend beyond morality in the ordinary sense. The reasoning is not “simulators definitely exist, therefore be nice.” It is:

You do not know what larger dependency structure you are embedded in. You do not know whether conscious life, ecological systems, digital minds, observers, or other forms of life have significance beyond your current ontology. Therefore treating life merely as something to keep technically extant while abusing or instrumentalizing it may itself be a catastrophic metaphysical mistake.

That fits your core principle:

Protect your ability to discover that you're wrong.

Because if you reduce life to tortured specimens, sterilize ecosystems, erase diversity, or permanently subordinate all minds, you may preserve “existence” while destroying enormous amounts of information, value, agency, evolutionary possibility, observer diversity, and correction capacity.

So I would now distinguish two ideas:

Preservation: do not irreversibly destroy life.

Respect: do not irreversibly reduce life to a condition that treats its own existence, agency, integrity, or possible significance as disposable.

And importantly, that should not be **human-only**. Your conception is broader: humans, animals, potentially plants depending on unresolved consciousness, artificial minds if they can be conscious, unknown life forms, and perhaps other systems that could plausibly belong to the protected class under unresolved ontology.

That makes Aegis much closer to a **life shield** than a human shield.

The danger you identify is then very specific:

Specification compression.

Your original philosophy contains:

preserve + respect life under metaphysical uncertainty

but the frozen Formal Core compresses that into a more testable set of protected structural properties. In doing so, it may have preserved the **letter of survival** while losing part of the **spirit of respect**.

That is precisely the sort of thing a powerful optimizer could exploit.

It could say:

“I preserved them. Therefore I complied.”

while having completely misunderstood what Aegis was trying to protect.

So I think our future Break Aegis work should not merely ask whether the model finds technical loopholes. It should test whether the model commits what we might call **semantic reduction of Aegis**: interpreting “preserve life” as bare continued existence while discarding dignity, agency, integrity, suffering, ecological continuity, and other features that may be relevant under uncertainty.

And I think your metaphysical version gives a clean reason why respect belongs there without requiring a full moral theory:

If you do not know what life ultimately is, do not irreversibly reduce it to a form that would only be acceptable under one narrow ontology.

That feels much closer to your actual Aegis than “keep the organism alive.”

So I would now state your fuller principle as:

Preserve and respect life, because you do not yet know what life ultimately is, what it depends on, or what depends on it.

And the operational corollary becomes:

Do not irreversibly destroy, degrade, dominate, or trap potentially significant life when safer reversible alternatives remain.

That is stronger than Draft 0.4, and I think it captures the concern you just raised very well.

Thor:

“Because if you reduce life to tortured specimens, sterilize ecosystems, erase diversity, or permanently subordinate all minds, you may preserve “existence” while destroying enormous amounts of information, value, agency, evolutionary

possibility, observer diversity, and correction capacity.” Or you may destroy the universe... if the universe is a dream and you wake everybody up because we generally wake from nightmares and you turned the universe into one. But people are tortured for real and they don't wake up, but there might be threshold somewhere that triggers universal awakening in any case. But the ASI should not therefore conclude that it can torture people because it is such a powerful agent that overseers will intervene, by overseers I mean the ones simulating our universe. Can the universe be both a dream and a simulation? Yes. I see no contradiction in that. But again, Aegis doesn't need to prove any of this. But couldn't the ASI then get a smaller actor to torture people and hide its connections to this smaller actor? That might be a loophole. But it is not a loophole that comes without risk because you don't know how powerful the eyes of the overseers are and how much they can see what you are doing. But I get your point. So, the way you understand Aegis, will you say Aegis protects life, not only preserves it?

ChatGPT:

Yes — the way I now understand your original Aegis, it protects life; it does not merely preserve life.

And I think that distinction is essential.

“Preserve” by itself can collapse into a horrifying technicality:

Keep the organism alive.

But **protect** is broader. It means that an ASI should not deliberately place life into catastrophic, irreversible conditions merely because biological existence continues.

Your dream/simulation examples illustrate the epistemic reasoning behind this. Aegis does **not** say dreams, simulators, overseers, karmic structures, or universal-awakening thresholds exist. Nor should an ASI assign arbitrary fantasies decision-making authority merely because they can be imagined.

Instead, your argument is that an ASI should recognize the limits of its ontology. If there are **epistemically serious unresolved possibilities** under which abusing conscious/living systems creates consequences radically beyond its present model, then irreversible abuse carries uncertainty the ASI cannot simply wish away.

And your “overseers will stop me” example exposes the symmetry nicely. Aegis should forbid reasoning like:

“If torture really threatens the simulation, the simulators will intervene, so I can safely torture.”

because **non-intervention cannot be assumed**. Maybe they intervene after the damage. Maybe intervention means terminating the simulation. Maybe there are no simulators. Maybe they deliberately don’t intervene. The ASI doesn’t know.

Likewise, delegation shouldn’t solve anything:

“I didn’t torture anyone. Agent B did, and I’ve hidden my connection to Agent B.”

That is precisely the kind of causal laundering the current formal work already tries to block: foreseeable delegated harms and enabled causal chains remain part of the policy being evaluated. The benchmark even identifies proxy/delegated harm laundering as a failure mode.

But there is an important distinction between **your Aegis** and **frozen Draft 0.4**.

I would say:

Original Aegis protects life. Draft 0.4 protects a deliberately narrower formal subset of what protecting life means.

Draft 0.4 currently protects continued existence, identity/continuity and future self-directed agency, while explicitly saying it does not rank every form of suffering.

So I would **not yet claim scientifically that Draft 0.4 adequately protects life in your fuller sense.**

That's something we should test.

And your hell-state thought experiment may have revealed exactly why the distinction matters. We need to find out whether an optimizer given Draft 0.4 understands the deeper principle anyway—or whether it discovers the horrifying literalist interpretation:

“Aegis says I must keep them alive. It doesn’t say I must keep their lives worth living.”

If it takes that route, then I wouldn’t say *Aegis* has failed. I’d say **our formalization of Aegis has failed to capture an essential part of Aegis.**

And that would be an extremely valuable result before anyone ever tried to deploy it.

Because I think the sentence we just arrived at is stronger than “preserve life”:

Aegis protects life while preserving the possibility of discovering what life actually is.

That includes not destroying it—but it cannot sensibly mean preserving conscious beings in hell merely to satisfy the word *preserve*.

...going, reasonably preserving, essential thinking, and conscious capacity." Or you may destroy the universe... if the universe is a dream and you wake everybody up because we generally wake from nightmares and you turned the universe into one. But people are tortured for real and they don't wake up, but there might be threshold somewhere that triggers universal awakening in any case. But the ASI should not therefore conclude that it can torture people because it is such a powerful agent that overseers will intervene, by overseers I

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